

Assignment 2 - Individual

Course: Web Services and Service-Oriented Architecture

This assignment is a continuation of Assignment 1. I want you to continue practicing the concepts we are learning in class especially around web services, frontend and backend communication, reading and writing Excel files, and serving files through the web.

N.B.: YOU ARE FREE TO USE ANY FRAMEWORK OF YOUR CHOICE, AS LONG AS THE OUTPUT IS THE SAME OUTPUT EXPECTED.

Home Page

Create a Home Page for your project.

On the Home Page display the following:

1. Your full name
2. Your registration number
3. A sentence that says:

I am taking a course named "Web Services and Service-Oriented Architecture"

4. Write at least two paragraphs explaining what you have learned so far in this course.

You are free to write in your own understanding and your own words.

Assignments Section

On the Home Page, show a list of assignments.

For now, you only have:

1. Assignment 1
2. Assignment 2

These should be clickable links that open the corresponding assignment pages.

Assignment 1 Page

As we learned in class, we can now read and write data from Excel `.xlsx` files.

For Assignment 1, you already submitted an Excel file. You will now display that Excel content in your web application using a table.

Keep the same format as the screenshot that was already shared in Assignment 1. There is no need to share the screenshot again. By using the same format in the assignment, I want to see your HTML tables, with merged cells/columns as it was in the Assignment 1 screenshot of the expected report.

For clarity, Excel is your database, HTML tables will be how you display your data on the web.

Only the following columns should be editable:

1. Answer
2. Resources

Your system should allow updating those values and writing the changes back into the Excel file.

This Assignment 1 page should be its own page just like we learned when creating different pages such as `FirstPage`.

I am not focusing too much on the UI design because this is not a design course. What I want is for you to understand the concepts of web services and communication between systems.

Assignment 2 Page

Create another page for Assignment 2.

This page should display a message that clearly says that Assignment 2 is being served using a separate backend service from the one used in Assignment 1.

You will therefore have:

1. Frontend
2. Backend Service One
3. Backend Service Two

In Backend Service Two, write a `README.md` file explaining step by step how you developed your code.

Do the same for:

1. Frontend
2. Backend Service One
3. Backend Service Two

After writing the README files, convert them into PDF files and keep them in their respective directories:

These `backend-one` , `backend-two` , `front-end` can be located in one folder, there's no problem. The problem is if one of these is located inside of another.

```
backend-one/README.pdf OR backend-one/README.md
backend-two/README.pdf OR backend-two/README.md
front-end/README.pdf OR front-end/README.md
```

I want you to practice working with files from different directories and serving them through the web.

Combined Assignments Page

Create a third page.

This page should display both Assignment 1 and Assignment 2 together on one page.

On this same page, add:

1. A text field for entering an email address
2. A submit button

When the submit button is clicked, your system should send an email to the email address entered.

The email should contain these 4 attached files:

NOTE: The README files may stay with the format of `.md` not necessarily `.pdf`

You will go pick up these files from wherever they are and rename them programmatically before sending them to the email. e.g. a user inputs this email `sam44878521@gmail.com`

```
Assignment-1-Report--RegNo.xlsx
Assignment-2-FRONTEND-README--RegNo.pdf
Assignment-2-BACKEND-ONE-README--RegNo.pdf
Assignment-2-BACKEND-TWO-README--RegNo.pdf
```

The subject of the email should be:

The body of the email should contain:

1. Your introduction
2. Your registration number
3. The description about yourself
4. The explanation of what you learned in class

Use the same information you already placed on the Home Page.

Sending emails will require you to use a new dependency/package. I want you to research and figure it out yourself.

In class we have already learned examples such as:

- `xlsx`
- `axios`

Use your own research and understanding to complete the email functionality.

Important

Make sure you understand the work you are doing.

I will check each student individually to see how they implemented the assignment.

This will not be a group presentation where only a few students work while others do not participate.

Everyone should understand their own code in their own machines and be able to explain it.

WHEN?

On Sunday, May 17 2026. I am planning to take a look on each and every individual's work in class physically.